CEFET/RJ

DAL & Comunidade

OBJETOS

Eduardo Ogasawara
eogasawara@ieee.org
https://eic.cefet-rj.br/~eogasawara

Objetos

- Variáveis armazenam objetos
  - Vetores
  - Atributos
  - Matrizes
  - Classes
  - Conversão
  - Listas
  - Tabelas

Exemplo

- Dados
  - 1, 2, 3, 4, 5, 6
- Cartas de Baralho
  - Naipes (Copas, Espadas, Ouros e Paus)
  - Face (Ás, 2, 3, 4, 5, 6, 7, 8, 9, 10, Valete, Dama e Rei)

https://rstudio-education.github.io/hopr/project-1-weighted-dice.html
https://rstudio-education.github.io/hopr/project-2-playing-cards.html

Vetores

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
dado <- c(1, 2, 3, 4, 5, 6)
dado
is.vector(dado)
```

Console

```
> dado <- c(1, 2, 3, 4, 5, 6)
> dado
[1] 1 2 3 4 5 6
> is.vector(dado)
[1] TRUE
>
```

Environment

dado	num [1:6] 1 2 3 4 5 6

Variáveis simples são vetores unitários

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```
numero <- 5
numero

is.vector(numero)

length(numero)

length(dado)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> numero
[1] 5
>
> is.vector(numero)
[1] TRUE
>
> length(numero)
[1] 1
>
> length(dado)
[1] 6
>
```

Environment History Connections Tutorial

Import | 107 MiB | List

R | Global Environment

values

dado	num [1:6] 1 2 3 4 5 6
numero	5

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Variável inteira e alfanumérica (strings)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
inteiro <- 1L
texto <- "ace"

typeof(inteiro)
typeof(texto)
```

Console

```
> inteiro <- 1L
> texto <- "ace"
>
> typeof(inteiro)
[1] "integer"
> typeof(texto)
[1] "character"
>
```

Environment

inteiro	1L
texto	"ace"

Vetores inteiros, alfanuméricos e funções aplicáveis

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
cartas <- 1L:13L
faces <- c("ás", "dois", "três", "quatro",
           "cinco", "seis", "sete", "oito",
           "nove", "dez", "valete", "dama", "rei")

n <- sum(cartas)
is.integer(n)

m <- max(faces)
m
```

1:1 (Top Level) R Script

Console

```
> cartas <- 1L:13L
> faces <- c("ás", "dois", "três", "quatro",
+            "cinco", "seis", "sete", "oito",
+            "nove", "dez", "valete", "dama", "rei")
>
> n <- sum(cartas)
> is.integer(n)
[1] TRUE
>
> m <- max(faces)
> m
[1] "valete"
>
```

Environment

values	
cartas	int [1:13] 1 2 3 4 5 6 7 8 9 ...
faces	chr [1:13] "ás" "dois" "três"...
m	"valete"
n	91L

Files

Name	Size	Modified
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Variável double

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```
1  dado <- c(1, 2, 3, 4, 5, 6)
2  dado
3  typeof(dado)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> dado <- c(1, 2, 3, 4, 5, 6)
> dado
[1] 1 2 3 4 5 6
> typeof(dado)
[1] "double"
>
```

Environment History Connections Tutorial

Import | 107 MiB | List

R | Global Environment

values

dado	num [1:6] 1 2 3 4 5 6

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Variável lógica

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```r
3 > 4

logico <- c(TRUE, FALSE,
            3 >= 4, 3 < 4,
            3 <= 4, 3 < 4,
            3 != 4, 4 == 4)
logico
typeof(logico)
```

9:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> 3 > 4
[1] FALSE
>
> logico <- c(TRUE, FALSE,
+             3 >= 4, 3 < 4,
+             3 <= 4, 3 < 4,
+             3 != 4, 4 == 4)
> logico
[1]  TRUE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
> typeof(logico)
[1] "logical"
>
```

Environment History Connections Tutorial

Import | 107 MiB | List

R | Global Environment

Values

logico	logi [1:8] TRUE FALSE FALSE T...

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Números complexos

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
comp <- c(1 + 1i, 1 + 2i, 1 + 3i)
comp

typeof(comp)
```

Console

```
> comp <- c(1 + 1i, 1 + 2i, 1 + 3i)
> comp
[1] 1+1i 1+2i 1+3i
>
> typeof(comp)
[1] "complex"
>
```

Environment

values	
comp	cplx [1:3] 1+1i 1+2i 1+3i

Variável raw (byte)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```
r <- raw(3)
typeof(r)

r[2] <- as.raw(255)
r[3] <- as.raw(1024)

r
```

Console

```
> r <- raw(3)
> typeof(r)
[1] "raw"
>
> r[2] <- as.raw(255)
> r[3] <- as.raw(1024)
Warning message:
out-of-range values treated as 0 in coercion to raw
>
> r
[1] 00 ff 00
>
```

Environment

r	raw [1:3] 00 ff 00

Atributos (metadados)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
dado <- c(1,2,3,4,5,6)
attributes(dado)

names(dado) <- c("um", "dois", "três",
                 'quatro', 'cinco', 'seis')
attributes(dado)
```

Console

```
> dado <- c(1,2,3,4,5,6)
> attributes(dado)
NULL
>
> names(dado) <- c("um", "dois", "três",
+                  'quatro', 'cinco', 'seis')
> attributes(dado)
$names
[1] "um"     "dois"   "três"   "quatro" "cinco"  "seis"

>
```

Environment

Values	
dado	Named num [1:6] 1 2 3 4 5 6

Files

Name	Size	Modified
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Retirando nomes

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```
1  names(dado) <- NULL
2  dado
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> names(dado) <- NULL
> dado
[1] 1 2 3 4 5 6
>
```

Environment History Connections Tutorial

Global Environment

values

dado num [1:6] 1 2 3 4 5 6

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Alterando as dimensões

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
dado <- 1:6
dim(dado) <- c(2, 3)
dado

attributes(dado)
```

Console

```
> dado <- 1:6
> dim(dado) <- c(2, 3)
> dado
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
> attributes(dado)
$dim
[1] 2 3

>
```

Environment

Data	
dado	int [1:2, 1:3] 1 2 3 4 5 6
Values	
comp	cplx [1:3] 1+1i 1+2i 1+3i

Files

Name	Size	Modified
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Matrizes

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
m <- matrix(dado, nrow = 2)
m

n <- matrix(dado, nrow = 2, byrow = TRUE)
n
```

Console

```
> m
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6
>
> n <- matrix(dado, nrow = 2, byrow = TRUE)
> n
     [,1] [,2] [,3]
[1,]    1    2    3
[2,]    4    5    6
>
```

Environment

Data	
dado	num [1:2, 1:3] 1 2 3 4 5 6
m	num [1:2, 1:3] 1 2 3 4 5 6
n	num [1:2, 1:3] 1 4 2 5 3 6

Files: C: > R > main

Name	Size	Modified
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Classes

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
typeof(dado)

class(dado)

attributes(dado)
```

Console

```
> typeof(dado)
[1] "integer"
>
> class(dado)
[1] "matrix" "array"
>
> attributes(dado)
$dim
[1] 2 3

>
```

Environment

Data	
dado	int [1:2, 1:3] 1 2 3 4 5 6
Values	
comp	cplx [1:3] 1+1i 1+2i 1+3i

Data-Hora (timestamp)

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
now <- Sys.time()
now

typeof(now)

class(now)
```

Console

```
> now <- Sys.time()
> now
[1] "2022-09-30 13:01:46 -03"
>
> typeof(now)
[1] "double"
>
> class(now)
[1] "POSIXct" "POSIXt"
>
```

Environment

Values	
now	2022-09-30 13:01:46 -03

Dados categóricos

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```r
genero <- factor(c("feminino", "masculino",
                   "feminino", "masculino"))

typeof(genero)

attributes(genero)
```

7:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> typeof(genero)
[1] "integer"
>
> attributes(genero)
$levels
[1] "feminino"  "masculino"

$class
[1] "factor"

>
```

Environment History Connections Tutorial

Import | 103 MiB | List

R | Global Environment

Values

genero	Factor w/ 2 levels "feminino"...

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Conversões de tipo

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
unclass(genero)

n <- unclass(genero)

as.integer(genero)

as.character(genero)
```

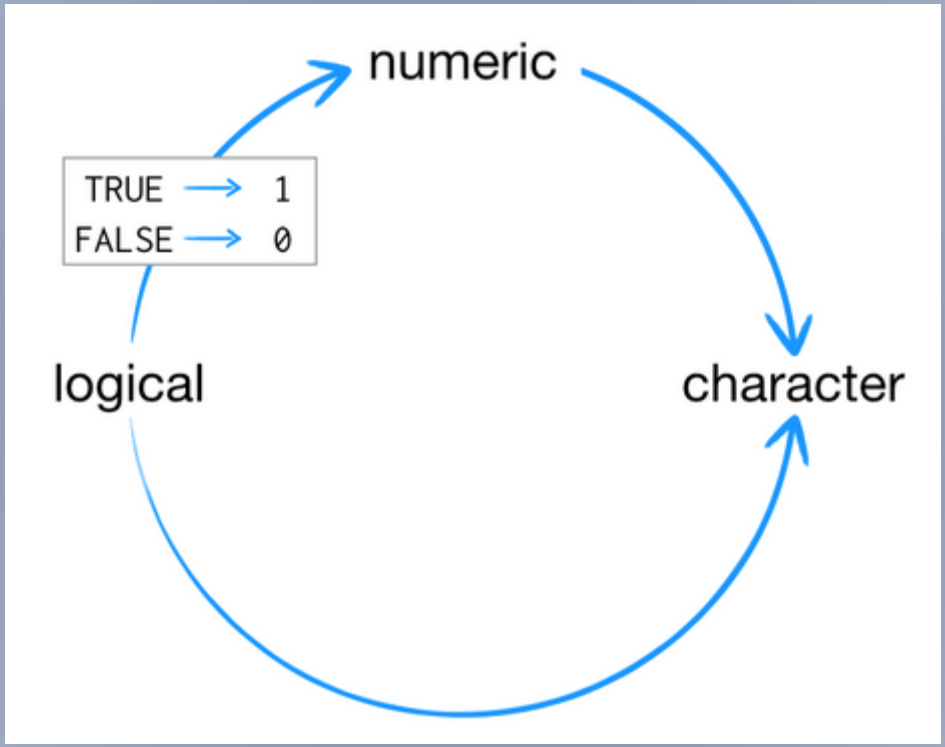
Console

```
> unclass(genero)
[1] 1 2 1 2
attr(,"levels")
[1] "feminino"  "masculino"
>
> n <- unclass(genero)
>
> as.integer(genero)
[1] 1 2 1 2
>
> as.character(genero)
[1] "feminino"  "masculino" "feminino"  "masculino"
```

Environment

values	
genero	Factor w/ 2 levels "feminino"...
n	int [1:4] 1 2 1 2
x	int [1:4] 1 2 1 2

Conversões



https://rstudio-education.github.io/hopr/r-objects.html#coercion

Mais sobre conversão

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | Project: (None)

exemplo.R*

Source on Save | Run | Source

```
1 sum(c(TRUE, TRUE, FALSE, FALSE))
2 sum(c(1, 1, 0, 0))
3 as.character(1)
4 as.logical(1)
5 as.numeric(FALSE)
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.2.1 · C:/R/main/

```
> sum(c(TRUE, TRUE, FALSE, FALSE))
[1] 2
> sum(c(1, 1, 0, 0))
[1] 2
> as.character(1)
[1] "1"
> as.logical(1)
[1] TRUE
> as.numeric(FALSE)
[1] 0
>
```

Environment History Connections Tutorial

Import | 104 MiB | List

R | Global Environment

Values

genero	Factor w/ 2 levels "feminino"...
n	int [1:4] 1 2 1 2
x	int [1:4] 1 2 1 2

Files Plots Packages Help Viewer Presentation

C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

tabelas

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
df <- data.frame(
    face = c("ás", "dois", "quatro"),
    naipe = c("ouros", "copas", "paus"),
    valor = c(1, 2, 4))
df
```

Console — R 4.2.1 · C:/R/main/

```
3 quatro  paus     4
> df <- data.frame(
+     face = c("ás", "dois", "quatro"),
+     naipe = c("ouros", "copas", "paus"),
+     valor = c(1, 2, 4))
> df
    face naipe valor
1     ás ouros     1
2   dois copas     2
3 quatro  paus     4
>
```

Environment — Global Environment

Data

df	3 obs. of 3 variables

Files — C: > R > main

Name	Size	Modified
..		
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,

Salvando e lendo arquivos

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

exemplo.R*

```r
write.csv(df, file = "cartas.csv",
          row.names = FALSE, quote=FALSE)

cartas <- read.csv("cartas.csv")
```

Console

```
> write.csv(df, file = "cartas.csv",
+           row.names = FALSE, quote=FALSE)
>
> cartas <- read.csv("cartas.csv")
>
```

Environment

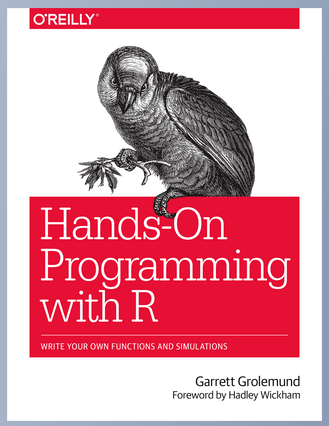
Data	
cartas	3 obs. of 3 variables
df	3 obs. of 3 variables

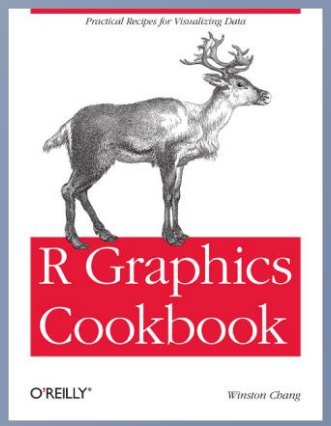
Files

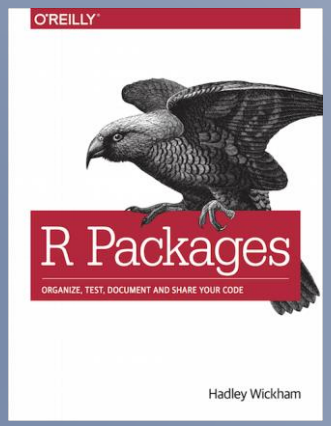
Name	Size	Modified
.RData	122 B	Sep 19,
.Rhistory	26.1 KB	Sep 28,
cartas.csv	60 B	Sep 30,

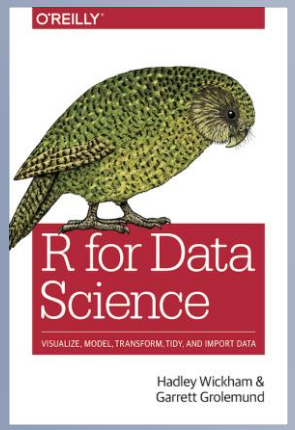
Referências

Código: https://nbviewer.org/github/eogasawara/R/blob/main/02-Pacotes-Ajuda.ipynb









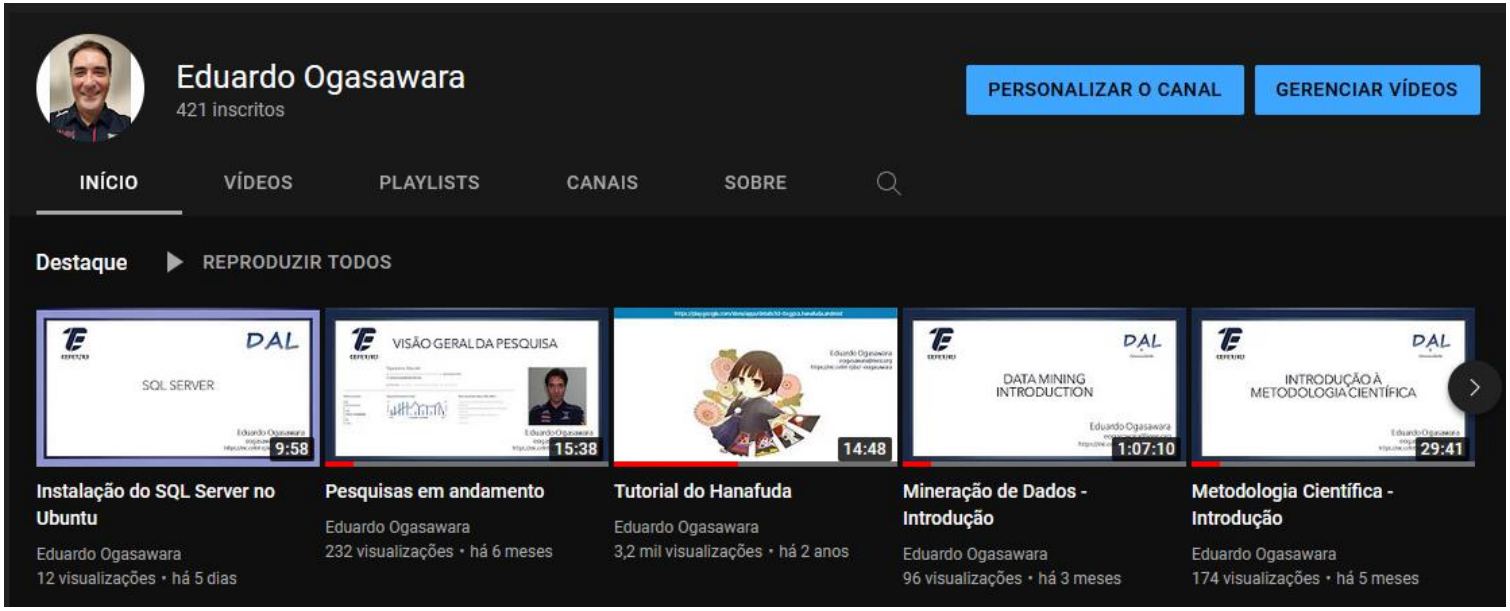
Hands-on Programming with R: https://rstudio-education.github.io/hopr/index.html
R Graphics Cookbook: https://r-graphics.org
R Packages: https://r-pkgs.org/index.html
R for Data Science: https://r4ds.had.co.nz

Leitura adicional: https://rstudio-education.github.io/hopr/r-objects.html

[1] G. Grolemund, 2014, Hands-On Programming with R: Write Your Own Functions and Simulations. O'Reilly Media, Inc.

Mantenha-se atualizado

Inscreva-se: https://eic.cefet-rj.br/~eogasawara/youtube



Slides and Jupyter notebooks: https://github.com/eogasawara/R